

Advanced (Habanero)

Q1. Simplify $\sqrt{16}$

- (A) 2
- (B) 4
- (C) 6
- (D) 8
- (E) 10

Q2. Convert $\sqrt[3]{8}$ to rational exponent form

- (A) $2^{1/2}$
- (B) $2^{1/3}$
- (C) 2^3
- (D) $8^{1/3}$
- (E) $2^{2/3}$

Q3. What is $\sqrt{25} \cdot \sqrt{4}$?

- (A) 10
- (B) 12
- (C) 15
- (D) 20
- (E) 30

Q4. Simplify $(\sqrt{9})^2$

- (A) 3
- (B) 6
- (C) 9
- (D) 12
- (E) 15

Q5. What is $\sqrt[3]{27}$?

- (A) 3
- (B) 6
- (C) 9
- (D) 12
- (E) 15

Q6. Simplify $\sqrt{36} \cdot \sqrt{9}$

- (A) 10
- (B) 12
- (C) 15
- (D) 18
- (E) 20

Q7. Convert $x^{1/2}$ to radical form

- (A) $\sqrt[3]{x}$
- (B) $\sqrt{x}$
- (C) x^2
- (D) x^3
- (E) $\sqrt[4]{x}$

Q8. Simplify $(\sqrt[3]{8})^3$

- (A) 2
- (B) 4
- (C) 6
- (D) 8
- (E) 10

Open Ended Questions (FRQ)

Q9. Simplify the expression $\sqrt{16} \cdot \sqrt[3]{8}$ and convert it to rational exponent form

Q10. Prove that $(\sqrt{x})^2 = x$ for all $x \geq 0$

Chef's Tips

- Remind students to simplify radicals before converting to rational exponent form
- Emphasize the importance of using the correct notation for radicals and rational exponents
- Encourage students to check their work by plugging in values for variables